

SPECIAL d20 Roleplaying Engine

Universal Edition

September 2026

A setting-neutral tabletop rules engine built around SPECIAL attributes, whole-number Skills, d20 Skill Tests, d10 Attribute Checks, tactical Action Points, meaningful equipment, and detailed burst-fire resolution.

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PART I - CORE SYSTEM

1. Dice Engine

The system uses two primary dice:

- **d20** for trained Skills.
- **d10** for direct SPECIAL checks and resistance checks.

The two dice have different jobs.

A Skill represents learned competence. A character with Skill 12 rolls 1d20 and succeeds on 1-12 before Difficulty is applied.

A SPECIAL attribute represents an underlying physical or mental capability. A direct attribute check uses 1d10 and succeeds on a result equal to or below the relevant attribute, with a natural 10 always failing.

1.1 Skill Test

Success Number = Skill - Difficulty

Roll 1d20.

If the result is equal to or below the Success Number, the test succeeds.

A result of 1 is always a particularly favorable result when the task allows an exceptional outcome. A result of 20 is always a failure unless a specific rule says otherwise.

1.2 Attribute Check

Roll 1d10 against the relevant attribute.

Roll ≤ Attribute = success

A natural **10 always fails**.

This remains true even if the attribute is 10 or higher.

An attribute of 10 therefore succeeds on 1-9. An attribute of 14 also succeeds on 1-9. Additional points above 10 remain valuable for formulas, secondary statistics, Skill formulas, damage, AP, and other effects without making a direct d10 check automatic.

1.3 Attribute Modifier

For rules that call for an attribute modifier:

Attribute Modifier = Attribute - 5

The modifier may be negative.

1.4 Rounding

Unless another rule specifies otherwise, round fractions to the nearest whole number. When a rule explicitly says to round down, always round toward zero.

2. SPECIAL

The seven core attributes are:

Attribute	Abbreviation	General Function
Strength	STR	Force, lifting, melee power, recoil control
Perception	PE	Awareness, precision, sensory information
Endurance	EN	Stamina, durability, recovery, environmental resistance
Charisma	CH	Social force, communication, leadership
Intelligence	IN	Reasoning, memory, technical ability, skill growth
Agility	AG	Coordination, speed, defense, action economy

Attribute	Abbreviation	General Function
Luck	LK	Chance, criticals, improbable outcomes

There is no universal maximum SPECIAL value.

The original 1-10 scale remains the foundation, but characters may rise above 10 through character creation, Backgrounds, Traits, Perks, equipment, or other campaign rules.

3. Secondary Statistics

3.1 Hit Points

At Level 1:

$$\text{HP} = 15 + \text{STR} + (2 \times \text{EN})$$

At each later level:

$$\text{HP gained} = 3 + \text{floor}(\text{EN} / 2)$$

Maximum HP is recalculated if permanent changes to STR or EN alter the character's maximum.

3.2 Armor Class

$$\text{Base AC} = \text{Armor AC} + \text{floor}(\text{AG} / 2) + \text{other modifiers}$$

The Agility contribution is deliberately smaller than the raw Agility score. A higher AC increases the Difficulty of attacks against the character.

Example: AG 8 contributes +4 AC. With Armor AC 2, the character has AC 6 before other modifiers.

3.3 Action Points

$$\text{AP} = 5 + \text{floor}(\text{AG} / 2)$$

AP is the resource used during combat to move, attack, reload, aim, brace, defend, use equipment, and perform other physical actions.

3.4 Carry Weight

$$\text{Carry Weight} = 25 + (25 \times \text{STR}) \text{ pounds}$$

Armor, equipment, perks, and setting-specific equipment can modify this value.

3.5 Melee Damage

For melee and unarmed attacks, add:

$$\text{Melee Damage Modifier} = \text{STR} - 5$$

The modifier may be negative.

A character with STR 3 therefore applies -2 to melee damage. A character with STR 12 applies +7.

A melee or unarmed weapon that would deal less than 1 damage before armor still deals at least 1 raw damage unless a rule specifically states that the attack can deal 0 damage.

3.6 Poison Resistance

$$\text{Poison Resistance} = \text{floor}(\text{EN} / 2)$$

Poison Resistance is a d10 target.

Roll 1d10 against the resistance value. Equal to or below the value succeeds. Natural 10 always fails.

Racial, Trait, Perk, armor, equipment, or environmental effects may add or subtract from the resistance target.

3.7 Radiation Resistance

$$\text{Radiation Resistance} = \text{floor}(\text{EN} / 5)$$

Radiation Resistance is also a d10 target.

Because EN commonly remains near the original SPECIAL scale, this produces a small but meaningful baseline. Armor and specialized equipment may provide much larger bonuses.

3.8 Sequence

$$\text{Sequence} = 2 \times \text{PE}$$

At the beginning of combat, the highest Sequence acts first. Ties are broken with a d10; higher result acts first.

3.9 Healing Rate

Use the following baseline:

EN	Natural Healing per Day
1-5	1 HP
6-8	2 HP
9-10	3 HP
11-12	4 HP
13-14	5 HP
15+	6 HP

A full uninterrupted day of rest doubles the amount recovered from ordinary natural healing.

4. Character Creation

Character creation combines random potential with meaningful choices.

4.1 Roll SPECIAL

Roll 1d10 seven times, once for each SPECIAL attribute. Then apply the **Creation Safety Net** before choosing a Background.

There is no hard cap of 10 after character creation.

Creation Safety Net

1. Replace every rolled 1 or 2 with 3.
2. Add the seven results. If the total is below 35, raise the lowest attributes one point at a time until the total reaches 35.
3. If the total is above 55, lower the highest attributes one point at a time until the total is 55, never reducing a rolled attribute below 3.

This keeps random generation while preventing an extreme opening array from producing a functionally unusable character or a single starting character with an excessive raw-stat advantage.

4.2 Choose a Background

Apply the Background after the initial SPECIAL rolls.

Every Background provides exactly **+5 SPECIAL points**, distributed according to that Background.

A Background may use any of the following patterns or another distribution totaling exactly five points:

- +1, +1, +1, +1, +1
- +2, +1, +1, +1
- +2, +2, +1
- +3, +1, +1
- +3, +2
- +4, +1
- +5

The distribution is part of the Background's identity. There is no universal requirement that every Background use the same pattern.

4.3 Spend 5 Free SPECIAL Points

After applying the Background, spend another **+5 SPECIAL points** as desired.

These points may be placed in any SPECIAL attributes and may raise an attribute above 10.

4.4 Choose Traits

Choose the normal number of Traits for the campaign.

Traits should substantially change how the character works rather than acting as small generic bonuses.

4.5 Choose Tag Skills

Choose three Tag Skills unless a Trait or Background changes the number.

Each Tag Skill receives +4 Skill at character creation.

4.6 Calculate Starting Skills

Use Appendix B to calculate starting Skills from the final SPECIAL values.

Apply Background adjustments, Traits, and Tag bonuses afterward.

4.7 Calculate Secondary Statistics

Calculate HP, AP, AC, Carry Weight, Melee Damage, Sequence, Poison Resistance, Radiation Resistance, and other derived values.

4.8 Finish the Character

Choose equipment, name, appearance, history, goals, relationships, and other elements appropriate to the campaign.

5. Backgrounds

A Background represents what the character has done and learned before the campaign begins.

A Background is not a species, class, or level.

Every Background grants exactly +5 SPECIAL points and **+5 starting Skill ranks**. The SPECIAL distribution is part of the Background package. Skill bonuses are listed explicitly as +2, +2, and +1. Backgrounds do not grant a second free pool.

Background	SPECIAL	Starting Skill Bonuses
Laborer	STR +3, EN +1, PE +1	Melee Weapons +2, Repair +2, Outdoorsman +1
Academic	IN +4, PE +1	Science +2, Doctor +2, First Aid +1
Soldier	STR +2, EN +2, PE +1	Small Guns +2, Big Guns +2, Outdoorsman +1
Criminal	AG +3, LK +1, PE +1	Sneak +2, Lockpick +2, Steal +1
Merchant	CH +2, LK +2, PE +1	Barter +2, Speech +2, Gambling +1
Rural	EN +2, PE +1, AG +1, IN +1	Outdoorsman +2, Repair +2, Melee Weapons +1
Technician	IN +3, AG +1, PE +1	Repair +2, Science +2, Small Guns +1
Performer	CH +4, LK +1	Speech +2, Gambling +2, Steal +1
Scout	PE +2, AG +2, EN +1	Sneak +2, Outdoorsman +2, Small Guns +1
Veteran	STR +2, EN +1, PE +1, LK +1	Small Guns +2, Big Guns +2, Repair +1

A GM may create as many Backgrounds as needed. The mechanical requirements are exactly +5 SPECIAL points and +5 starting Skill ranks.

6. Traits

Traits are broad rules that create meaningful differences between characters.

Finesse

Increase the character's Critical Range by 1 number.

Reduce melee and unarmed damage by 4.

Final melee or unarmed damage cannot be reduced below 1 by Finesse alone.

Gifted

Increase every SPECIAL attribute by 1.

Reduce every Skill by 2 at character creation.
Gifted does not reduce Skill Points gained at later levels.

Skilled

Increase every Skill by 2 at character creation.
Gain +1 additional Skill Point each level.
Gain a Perk every 4 levels instead of every 3 levels.

Jinxed

The character's Critical Failure Range is 18-20.
While the Jinxed character is in a fight, every combatant uses a Critical Failure Range of 18-20.

Lifegiver

Lifegiver has one rank.
Increase maximum HP by 4 for every character level, including levels already gained when the character acquires the Perk or effect.
A 6th-level character who gains Lifegiver immediately gains +24 maximum HP.

7. Skills

Skills are whole-number ratings.
The standard skill scale is 0-30.

Skill	General Use
Small Guns	Pistols, SMGs, rifles, shotguns, conventional firearms
Big Guns	Heavy machine guns, launchers, flamethrowers, support weapons
Energy Weapons	Directed-energy weapons and analogous advanced systems
Unarmed	Fists, kicks, unarmed techniques, powered hand weapons
Melee Weapons	Knives, clubs, swords, axes, polearms, and similar weapons
Throwing	Thrown weapons and explosives
First Aid	Basic treatment and stabilization
Doctor	Serious injuries, complex medicine, surgery
Sneak	Moving quietly and avoiding notice
Lockpick	Mechanical locks and bypass work
Steal	Pickpocketing, planting objects, theft
Traps	Detecting, setting, disarming, and using traps
Science	Analysis, research, advanced systems, theoretical knowledge
Repair	Mechanical and technical maintenance
Pilot	Vehicles, aircraft, and other complex transport
Speech	Persuasion, deception, intimidation, negotiation
Barter	Trade, appraisal, prices, exchange
Gambling	Games of chance, risk, tells, cheating detection

Skill	General Use
Outdoorsman	Tracking, navigation, food, water, field survival

A setting may rename or replace individual Skills without changing the core engine.

8. Skill Advancement, Tags, and Perks

8.1 Skill Costs

Current Skill	Cost for the next +1
1-10	1 Skill Point
11-20	2 Skill Points
21-25	3 Skill Points
26-30	4 Skill Points

Ordinary advancement cannot raise a Skill above 30 unless the GM expands the engine.

8.2 Skill Points per Level

Skill Points per Level = 5 + floor(IN / 2)

Examples:

IN	Skill Points per Level
1	5
5	7
8	9
10	10
12	11
15	12
20	15

8.3 Tag Skills

Each Tag Skill receives +4 Skill at creation.

When advancing a Tag Skill, halve the normal Skill Point cost and round up.

8.4 Perks

Unless a Trait, Perk, or campaign rule changes the schedule, gain one Perk every 3 levels.

PART II - RESOLUTION AND COMBAT

9. Skill Tests and Difficulty

9.1 Difficulty Scale

Difficulty	Meaning
-6	Almost trivial
-4	Very easy
-2	Easy
0	Normal
+2	Difficult
+4	Very difficult
+6	Extreme
+8	Exceptional
+10	Near-impossible

Difficulty modifies the Success Number directly.

Example:

Skill 14, Difficulty +4:

Success Number = 14 - 4 = 10

Roll 1d20. Results 1-10 succeed.

9.2 Impossible Tasks

If the Success Number is 0 or lower, the action normally cannot succeed through ordinary Skill use.

The GM may allow a special circumstance, tool, Perk, or unusual method to create a chance.

9.3 Degrees of Success

The amount by which a roll succeeds may matter.

Margin of Success = Success Number - die result

A margin of 5 or more indicates an especially strong success when the task allows such an outcome.

A natural 1 may produce an exceptional success when the task permits it.

10. Attribute Checks and Resistance

Direct checks against physical or mental capability use 1d10.

Roll <= Attribute succeeds.

Natural 10 always fails.

10.1 Strength Check

Use Strength for lifting, forcing, grappling, overpowering, and recoil control.

10.2 Agility Check

Use Agility for balance, reflexes, fast physical reactions, and tasks where coordination matters more than training.

10.3 Endurance Check

Use Endurance for exhaustion, sustained physical strain, resisting shock, toxins, environmental stress, and similar effects.

10.4 Perception Check

Use Perception to notice sensory details, hidden objects, distant movement, and sudden threats.

10.5 Intelligence Check

Use Intelligence for memory, reasoning, deduction, and tasks where a Skill does not apply.

10.6 Charisma Check

Use Charisma for broad social presence when no specific social Skill is appropriate.

10.7 Luck Check

Use Luck when the character has a genuine question of chance rather than a problem of learned competence.

10.8 Resistance Checks

Poison and Radiation Resistance use the same d10 roll-under format.

Armor may also supply Resistance values for Environmental hazards or other specific effects.

11. Awareness and Detection

Awareness may be passive or active.

11.1 Passive Detection

Use Perception plus situational Difficulty when the character is not actively searching.

11.2 Active Search

Use the appropriate Skill or Perception check when the character deliberately searches.

11.3 Visibility

Visibility is a situational condition rather than a Skill.

A GM may apply Difficulty for darkness, smoke, weather, camouflage, distance, concealment, or visual obstruction.

12. Time and Travel

The rules deliberately avoid unnecessary bookkeeping.

Use whatever level of time tracking the scene actually needs.

A combat round is approximately 10 seconds.

A brief exploration interval is approximately 1 minute.

A practical exploration block is approximately 10 minutes.

Long travel should track food, water, fuel, ammunition, fatigue, weather, and encounter pressure only when those resources create meaningful decisions.

13. Combat Basics

Combat uses rounds and turns.

At the start of combat, determine Sequence.

Each combatant receives AP each round according to their Agility. AP can be spent on movement, attacks, equipment, positioning, defense, and other actions.

A round ends after all combatants have completed their turns.

Surprise

Determine normal Sequence for everyone. Characters who are not surprised take their turns first in Sequence order. Surprised characters then act in normal Sequence order after all unsurprised characters have acted. Surprise applies only to the opening round; once a surprised character has taken that delayed first turn, normal Sequence resumes.

14. Actions and Action Points

Common actions:

Action	AP
Move 1 open-ground hex	1
Use an item already in hand	3
Draw or equip an item	4
Reload most firearms	2
Stand up	4
Change stance	2
Lay or deploy a device	6
Aim	2
Defend	Remaining AP

Weapon attacks use the AP cost listed by the weapon.

A character may split AP among several compatible actions.

15. Movement, Positions, and Cover

15.1 Movement

Open ground costs 1 AP per hex.

Rough ground generally costs 2 AP per hex.

Very difficult terrain may cost 3 AP or more.

15.2 Crouching

Changing from standing to crouching costs 2 AP.

When the position meaningfully protects the character, attacks against a crouching character suffer Difficulty +2. Crouching movement costs 2 AP per hex.

15.3 Prone

Going prone costs 2 AP.

A prone character gains Difficulty -2 on compatible ranged attacks. Movement while prone costs 4 AP per hex.

15.4 Cover

Cover	Attack Difficulty
Light cover	+2
Solid cover	+4
Near-complete cover	+6
Complete cover	Attack normally impossible

The GM should use the smallest meaningful set of modifiers rather than stacking tiny penalties.

16. Attacks

16.1 Attack Number

Start with the appropriate Skill.

Apply Difficulty from range, lighting, cover, armor, position, target size, weapon condition, and other circumstances.

Success Number = Skill - total Difficulty

Roll 1d20.

Equal to or below the Success Number succeeds.

16.2 Range

Weapons list effective Range in meters.

The GM may assign increasing Difficulty beyond the weapon's listed range. Use +2 Difficulty per additional range band rather than stacking +1 modifiers.

16.3 Lighting

Condition	Difficulty
Clear / well-lit	0
Light obstruction	+2
Medium obstruction	+4
Dark interior	+4
Heavy rain or dust	+6
Very dark	+8
Total darkness	Normally impossible without an appropriate sense or light source

16.4 Targeted Attacks

Targeted attacks trade accuracy for a specific body location.

Location	Additional Difficulty
Legs	+4
Arms	+6
Groin	+6
Head	+8
Eyes	+12

A successful attack that is not explicitly targeted strikes **Center Mass / Torso** by default. A Targeted Attack uses the declared location and its listed Difficulty.

A successful Critical Hit should use the targeted body location whenever the anatomy permits it.

17. Burst Fire

Burst fire is intended to be one of the system's detailed combat subsystems.

A burst is not simply a normal attack with a larger damage number. The system distinguishes:

- whether the shooter hits the target;
- whether the shooter controls recoil;
- how accurately the weapon holds the burst together; and
- where the burst was aimed.

17.1 Weapon Burst Statistics

Every burst-capable firearm may have:

Statistic	Function
Burst	Maximum number of rounds in one burst
Accuracy	Burst effectiveness; not an attack bonus
Recoil	Difficulty of physically controlling the burst
Burst AP	AP cost to fire a burst

17.2 Declare the Burst

Before rolling, the shooter declares:

1. Burst size, up to the weapon's listed Burst value.
2. Target Zone.
3. Whether the shooter is Bracing.

17.3 Make the Attack Roll

Make the normal attack roll.

If it fails, the burst does not hit.

If it succeeds, record the **Attack Margin**:

Attack Margin = Success Number - attack die

17.4 Recoil Training

Firearm skill improves recoil control.

For every full 5 points of the relevant firearm Skill above 5, reduce effective Recoil by 2.

Firearm Skill	Recoil Reduction
5 or less	0
10-14	2
15-19	4
20-24	6
25-29	8
30	10

This is a reduction to effective Recoil, not a bonus to the attack roll.

17.5 Recoil Check

Calculate:

Effective Recoil = Weapon Recoil - Recoil Training

Effective Recoil cannot be reduced below 0.

Then calculate:

Recoil Target = Effective STR - Effective Recoil

where Effective STR is normally the character's Strength.

The Recoil Target cannot be lower than 1 and cannot be higher than 9 for the purpose of a d10 Recoil Check.

Roll 1d10.

Roll <= Recoil Target = controlled burst

Natural 10 always fails.

17.6 Brace

Brace is a firing technique, not a generic attack bonus.

When a character fires while Braced:

- Double the normal AP cost of the attack.
- Double the character's Strength for the Recoil Check only.

A character may fire and Brace in the same round even if the doubled AP cost exceeds the character's normal maximum AP. The character cannot exceed their available AP through unrelated actions.

Example:

STR 5, Small Guns 20, weapon Recoil 10.

Small Guns 20 provides -6 Recoil.

Effective Recoil:

10 - 6 = 4

While Braced, effective STR is:

5 x 2 = 10

Recoil Target:

$$10 - 4 = 6$$

The shooter succeeds on 1-6 on the d10.

17.7 Recoil Control Margin

If the Recoil Check succeeds:

$$\text{Control Margin} = \text{Recoil Target} - \text{Recoil die}$$

If the Recoil Check fails, Control Margin is 0.

A failed Recoil Check does not automatically turn a successful attack into a miss. It represents a poorly controlled burst and reduces the number of rounds that can effectively land.

17.8 Accuracy

Accuracy is not added to the attack roll.

Accuracy improves Burst Effectiveness after the attack succeeds.

Typical firearms should have Accuracy from 0 to 3. Precision-built weapons and unusual configurations may reach 4 or 5.

17.9 Burst Effectiveness

Calculate:

$$\text{Burst Score} = (\text{Attack Margin} \times 2) + \text{Accuracy} + (\text{Control Margin} \times 2)$$

Then apply the Target Zone modifier.

Target Zone	Additional Burst Requirement
Center Mass	+0
Arms / Legs / Groin	+4
Head	+8
Eyes	+12

Subtract the Target Zone requirement from Burst Score.

17.10 Burst Hit Table

Effective Burst Score	Hits
0-6	1
7-9	2
10-12	3
13-15	4
16-18	5
19-21	6
22-24	7
25-27	8
28-30	9
31+ Full burst, up to the weapon's declared burst size	

The result is capped at the number of rounds actually fired.

There is no hard target-location cap. Center mass, the head, and even the eyes can theoretically receive the entire declared burst if the shooter and weapon produce a sufficiently high Burst Score.

17.11 Where the Rounds Land

For a burst aimed at a body area, the first successful hit is assumed to strike the declared zone.

Additional hits follow the same general pattern but may spread around the declared zone.

Use these defaults:

Target Zone	Additional Hits
Center Mass	Torso first, then arms, legs, groin, or nearby cover/targets as appropriate
Arms	Primarily the declared arm, with spread into torso or shoulder
Legs	Primarily the declared leg, with spread into hips or torso
Groin	Primarily groin/upper-leg region, with spread into torso or legs
Head	Primarily head, with spread into neck or torso
Eyes	Primarily the declared eye, with spread into the head

The GM may roll a location for each hit when the exact distribution matters.

17.12 Burst Example

A shooter has Small Guns 18, STR 7, and uses a weapon with:

Burst 10

Accuracy +2

Recoil 8

The shooter fires a 10-round center-mass burst.

Their attack Success Number is 14 and they roll a 6.

Attack Margin:

$$14 - 6 = 8$$

Small Guns 18 gives Recoil -4.

Effective Recoil:

$$8 - 4 = 4$$

Unbraced Effective STR is 7.

Recoil Target:

$$7 - 4 = 3$$

The shooter rolls a 2 on the Recoil Check.

Control Margin:

$$3 - 2 = 1$$

Burst Score:

$$(8 \times 2) + 2 + (1 \times 2) = 20$$

Center Mass has no additional requirement.

A Score of 20 produces **6 hits**.

Each of those six rounds receives its own damage resolution.

If the same shooter Braced, Effective STR would become 14. The Recoil Target would rise to 10 but is capped at 9, making the shooter much more likely to produce a tightly controlled burst.

17.13 Burst Damage

Fast Burst - Default

Roll damage once and multiply the result by the number of hits. Apply armor and other damage rules after the multiplication. If the attack roll is a Critical Hit, apply one Critical Hit effect to the burst.

Per-Bullet Burst - Optional Module

Resolve each hit separately, including a separate damage roll for each bullet. Use this module only when ammunition expenditure, individual wounds, or hit-by-hit resolution is important. A burst still receives at most one Critical Hit from the attack roll unless a rule specifically allows additional criticals.

18. Damage, Armor, and Penetration

18.1 Damage Types

Use four broad damage categories.

Type	Includes
Normal	Unarmed attacks, melee weapons, claws, bites, blunt trauma, ordinary physical attacks
Ballistic	Firearms, bullets, shot, fragmentation, kinetic projectiles
Energy	Lasers, plasma, focused energy, other directed-energy attacks
Environmental	Radiation, fire, gas, poison exposure, corrosive chemicals, extreme environmental hazards

The categories are deliberately broad.

A campaign does not need separate armor entries for every individual chemical, heat source, gas, or radiation source unless a special effect specifically requires one.

18.2 Damage Threshold

DT is the amount of damage armor absorbs before Damage Resistance is applied.

If Raw Damage is less than or equal to DT, the attack deals no damage unless an effect specifically bypasses DT.

Damage after DT = $\max(0, \text{Raw Damage} - \text{DT})$

18.3 Damage Resistance

DR is a percentage applied to the remaining damage after DT.

For example, DR 20% reduces 20 points to 16.

Final Damage = Damage after DT - $\text{floor}(\text{Damage after DT} \times \text{DR})$

DR can therefore remain a percentage even though Skills no longer use percentage notation.

18.4 Armor Penetration

PEN means Armor Penetration.

Use PEN instead of AP so it cannot be confused with Action Points.

PEN reduces DT and may also reduce DR when a specific ammunition or effect says so.

Adjusted DT = $\max(0, \text{DT} - \text{PEN})$

The standard low-grade armor-piercing ammunition rule is:

- **PEN 6**
- **DR is halved, rounded down**
- **Damage -1**

More advanced ammunition may have higher PEN values and stronger effects.

Example:

A target has:

DT 10 / DR 40%

An AP round with PEN 6 changes the armor to:

DT 4 / DR 20%

Then normal damage is resolved.

18.5 Damage Example

A rifle deals 18 Ballistic damage.

The target has DT 10 and DR 40%.

After DT:

$$18 - 10 = 8$$

Damage Resistance removes:

$$\text{floor}(8 \times 0.40) = 3$$

Final damage:

5

The same attack using PEN 6 ammunition gives:

DT 10 -> 4

DR 40% -> 20%

After DT:

$$18 - 4 = 14$$

DR removes 2.

Final damage:

12

18.5 Combat Conditions

0 HP / Downed. At 0 HP, a character becomes Downed. A Downed character cannot spend AP or perform ordinary actions and cannot Defend. They remain in their current position until stabilized and raised above 0 HP, or until they die. Further damage to a Downed character is handled normally and may cause additional conditions.

Bleeding. A Bleeding character loses 1 HP at the end of each of their turns. If the character reaches 0 HP while Bleeding, they remain Downed and continue to bleed. A successful First Aid or Doctor treatment ends Bleeding. Bleeding from multiple wounds does not stack unless a weapon, effect, or critical rule explicitly says it does.

Crippled. A Crippled body part suffers a lasting functional penalty. A crippled arm cannot be used normally for attacks requiring that arm; a crippled leg doubles movement AP and prevents normal sprinting; other locations use the penalty listed by the relevant Critical Hit table. A Crippled condition remains until treated by an appropriate medical action or campaign effect.

Dead. A character dies when a lethal effect explicitly causes death, when an injury is ruled immediately fatal, or when the character reaches **-5 HP**. The -5 threshold is the standard human-scale default and may be changed by the campaign. Until the death threshold is reached, a character below 0 HP remains Downed and may be stabilized.

19. Critical Hits and Critical Failures

19.1 Critical Range

Critical Hits occupy the **low end** of the d20. A Critical Hit is possible only after a successful attack. Natural 20 remains a failure and is never a Critical Hit.

Luck	Critical Range
1-5	1
6-10	1-2
11-15	1-3
16-20	1-4
21-25	1-5
26-30	1-6

Continue the pattern for higher Luck.

Finesse expands the Critical Range by 1 number at the low end. For example, Luck 10 with Finesse has a Critical Range of 1-3.

19.2 Critical Hit Procedure

After a successful attack, check the natural d20 result against the attacker's Critical Range. If it falls inside the range, the attack is a Critical Hit. Resolve the normal damage, then apply one appropriate Critical Hit effect. When a location was targeted, use that location's Critical Hit table; otherwise use Center Mass/Torso or another applicable general location.

A natural 20 is always a failure on a Skill Test. It is not a Critical Hit.

19.3 Head Criticals - d20

d20	Effect
1-2	Severe pain; Difficulty +2 on actions until next turn
3-4	Knockdown
5-6	Bleeding wound
7-8	Concussion; Difficulty +2 on mental checks for 1d4 rounds
9-10	Double damage
11-12	Knockdown and double damage
13-14	Severe wound; Stunned for 1 round
15-16	Unconscious for 1d4 rounds
17-18	Armor bypass for this hit and double damage
19	Devastating head wound; immediate death only when plausible
20	Catastrophic head injury; use the most severe plausible effect

19.4 Eye Criticals - d20

d20	Effect
1-4	Eye injury; sight-based checks suffer Difficulty +4 until treated
5-8	Severe facial bleeding; attack Difficulty +2
9-12	Disorientation; lose 2 AP next turn
13-16	Eye blinded until medically treated
17-18	Blindness and double damage

d20	Effect
19	Severe facial injury
20	Catastrophic eye/head injury

19.5 Torso Criticals - d20

d20	Effect
1-4	Bleeding wound
5-8	Knockdown
9-12	+5 damage
13-16	Double damage
17-18	Armor bypass and double damage
19	Severe internal injury
20	Catastrophic torso injury

19.6 Arm Criticals - d20

d20	Effect
1-4	Arm pain; Difficulty +2 using the limb
5-8	Drop held item
9-12	Bleeding wound
13-16	Arm crippled
17-18	Double damage and arm crippled
19	Arm rendered unusable
20	Severe fracture or amputation when plausible

19.7 Leg Criticals - d20

d20	Effect
1-4	Difficulty +2 on movement-related checks
5-8	Fall prone
9-12	Movement reduced until treated
13-16	Leg crippled
17-18	Double damage and leg crippled
19	Leg rendered unusable
20	Severe fracture or amputation when plausible

19.8 Groin Criticals - d20

d20	Effect
1-4	Severe pain; lose 2 AP next turn
5-8	Knockdown
9-12	Stunned for 1 round
13-16	Bleeding and severe pain
17-18	Double damage and Stunned
19	Severe disabling injury
20	Catastrophic injury appropriate to the target

19.9 Critical Failures

A normal character has a Critical Failure Range of natural 20 on attacks. A natural 20 is always a failure; if it falls in the character's Critical Failure Range, also apply a Critical Failure effect.

Jinxed expands the Critical Failure Range to 18-20 for the character and every combatant in the fight.

The Critical Failure system is separate from the Critical Hit system: Critical Hits use the low end of the d20; Critical Failures use the high end.

Melee Critical Failures - d20

d20	Result
1-2	Weapon slips; lose 1 AP
3-4	Lose remaining attack AP
5-6	Hit the nearest valid secondary target if one exists
7-8	Half normal weapon damage to yourself
9-10	Fall prone
11-12	Drop weapon
13-14	Hand or arm injured; Difficulty +2 with that weapon until treated
15-16	Weapon damaged
17	Weapon breaks
18	Full weapon damage to yourself
19	Full weapon damage to yourself and knockdown
20	Catastrophic melee mishap; GM chooses a severe plausible consequence

Ranged Critical Failures - d20

d20	Result
1-2	Misfeed or failure to cycle; clear the weapon
3-4	Ammunition problem; lose 1 AP clearing it
5-6	Lose remaining attack AP
7-8	Shot strikes the nearest valid secondary target if one exists
9-10	Weapon shifts; next attack suffers Difficulty +2
11-12	Shooter falls prone
13-14	Shooter suffers 1d4 damage from weapon or environment
15-16	Weapon damaged
17	Magazine or feed problem; reload required
18	Weapon breaks
19	Accidental discharge against a valid nearby target
20	Catastrophic firearm malfunction; GM chooses a severe plausible consequence

20. Special Combat Rules

20.1 Knockdown

Heavy melee attacks, explosive force, and sustained bursts can knock a target down.

A GM may require an Endurance Check after:

- a heavy melee impact;
- a blast or explosive effect;
- a burst that lands more than half of its possible hits;
- a weapon or effect explicitly designed to knock targets down.

20.2 Getting Up

Getting up costs 4 AP.

A prone character loses the Agility portion of AC until upright unless a specific effect says otherwise.

20.3 Aiming

Aiming costs 2 AP. The next compatible ranged attack gains Difficulty -2. A character may aim twice while remaining stationary, reaching a total Difficulty -4.

Aiming is lost if the character moves significantly or takes an incompatible action.

20.4 Defending

A character who deliberately ends their turn without spending all AP may defend.

Add the unused AP to AC until the beginning of their next turn, to a maximum **+4 AC** from Defend. This bonus is not saved between rounds.

PART III - SKILLS IN PLAY

21. Medicine and Technical Tasks

21.1 First Aid

Combat AP: 6. The user must be adjacent to the target and able to reach the injury. Make a First Aid Skill Test. On success, restore 1d10 HP and end the Bleeding condition; a character at 0 HP becomes able to act once they have sufficient HP to do so. On failure, the 6 AP are spent and the target receives no benefit. Outside combat, First Aid can be performed over several minutes.

21.2 Doctor

Doctor handles serious injury, severe bleeding, fractures, crippled limbs, surgery, and other conditions beyond ordinary First Aid.

A successful treatment may restore 2d10 HP over roughly one hour outside combat when appropriate equipment is available.

21.3 Science

Science answers questions about how systems work, what a device does, what a sample contains, and how unfamiliar technology should be interpreted.

21.4 Repair

Repair restores, modifies, sabotages, and maintains physical equipment.

When a task requires both theoretical understanding and physical intervention, Science and Repair may be used together.

21.5 Pilot

Pilot covers operation under pressure, evasive maneuvers, navigation through difficult conditions, and keeping a vehicle under control during dangerous operation.

22. Stealth, Locks, Theft, and Traps

22.1 Sneak

Sneak is normally rolled when meaningful hidden movement begins and when circumstances change significantly.

Armor, noise, lighting, terrain, and observation can modify Difficulty.

22.2 Lockpick

Lockpick handles mechanical locks, safes, bypass devices, and similar systems.

Time and tools may reduce Difficulty.

22.3 Steal

Steal covers taking objects without immediate detection, planting objects, and similar acts of manual theft.

22.4 Traps

Traps covers detection, disarming, construction, and deliberate use.

A trap's Difficulty should reflect complexity, concealment, time, and available tools.

23. Social Skills

23.1 Speech

Speech covers persuasion, intimidation, deception, negotiation, leadership, and social pressure.

Different personalities may produce different fictional approaches without changing the core mechanic.

23.2 Barter

Barter covers prices, exchange, appraisal, trade relationships, and economic negotiation.

23.3 Gambling

Gambling covers formal games of chance, reading tells, recognizing suspicious play, and managing risk.

24. Survival and Exploration

24.1 Outdoorsman

Outdoorsman covers navigation, tracking, food, water, weather awareness, identifying hazards, campcraft, and general field survival.

The GM should distinguish between knowledge and pure endurance. A character may know how to cross difficult terrain without personally being good at enduring the trip.

PART IV - CHARACTER GROWTH

25. Experience and Levels

Award Experience for meaningful accomplishments, discoveries, successful missions, difficult challenges, and other campaign goals.

The exact Experience schedule can vary by campaign.

When a character gains a level:

1. Increase maximum HP by the normal level amount.
2. Gain Skill Points.
3. Increase Skills or retain points for later spending.
4. Gain a Perk when the character reaches the campaign's Perk interval.
5. Recalculate any derived statistics affected by permanent changes.

26. Perks

Perks are significant abilities that modify the rules. Unless a campaign changes the schedule, gain one Perk every 3 levels.

Perk	Effect
Action Boy / Action Girl	+1 AP per rank; maximum 2 ranks.
Bonus HtH Damage	+2 Melee Damage per rank; maximum 3 ranks.
Bonus Ranged Damage	+2 ranged damage per rank on successful hits.
Bonus Move	The first 2 movement hexes each turn cost 0 AP; each rank adds 2 more free hexes.
Bracing	Gain Brace benefits without doubling AP for eligible supported firing positions.
Educated	+2 Skill Points per level per rank; maximum 3 ranks.
Lifegiver	+4 maximum HP per rank.
Quick Pockets	Reduce the AP cost of drawing or equipping an item by 2.
Sharpshooter	+2 effective Perception for ranged distance calculations.
Strong Back	+50 lb Carry Weight per rank.
Survivalist	+4 Outdoorsman per rank.
Thief	+2 Sneak, Lockpick, Steal, and Traps.
Toughness	+5% Damage Resistance to ordinary damage categories.
Weapon Handling	+3 effective Strength for weapon Strength requirements and recoil.
Better Criticals	Roll twice on a Critical Hit table and use the more severe result.

Perk	Effect
Awareness	Learn an observed target's approximate HP band, armor class band, and weapon category.
Quick Hands	Reduce reload AP by 1, minimum 1.
Iron Grip	Reduce firearm Recoil by 2 while Braced.
Field Medic	First Aid costs 4 AP instead of 6.
Steady Aim	The first Aim action each turn adds another Difficulty -2 to the next compatible ranged attack.

Perks should be stronger and more specific than a Background.

27. Training and Other Skill Sources

Training can provide Skills outside level advancement.

Books, instructors, equipment, repeated practice, formal education, or campaign milestones may grant Skill increases.

A GM should distinguish temporary instruction from a permanent improvement in competence.

PART V - EQUIPMENT FRAMEWORK

28. Weapons

Every weapon should list:

Field	Meaning
Damage	Raw damage before armor
Type	Normal, Ballistic, Energy, or Environmental
Range	Effective range in meters
AP	AP cost for the attack
Targeted AP	AP cost for a targeted attack
Burst	Maximum burst size, if applicable
Accuracy	Burst effectiveness rating
Recoil	Burst control difficulty
PEN	Armor penetration when inherent to the weapon
Capacity	Ammunition carried
Weight	Weapon weight
Min STR	Strength needed to use it comfortably

28.1 Generic Melee Examples

Weapon	Damage	Type	AP	Targeted AP	Min STR	Weight
Knife	1d8 + MD	Normal	3	4	2	2 lb
Club	1d10 + MD	Normal	3	4	3	4 lb
Sword	1d12 + MD	Normal	4	5	4	4 lb
Spear	1d10 + MD	Normal	3	4	3	5 lb
Heavy Hammer	2d8 + MD	Normal	5	6	6	12 lb

29. Firearms and Weapon Construction

Firearm construction uses parts rather than a giant list of unrelated weapon-specific exceptions.

A firearm's final profile is assembled from:

Cartridge + action + barrel + stock + sights + magazine + recoil system + optional special parts

The weapon's final Damage, Range, AP, Accuracy, Recoil, Capacity, Weight, and Minimum STR are the sum of those component effects.

29.1 Base Firearm Families

Family	Typical Range	Typical AP	Typical Burst
Pistol	15-30 m	4-6	None or 3-10
SMG	20-35 m	5-6	3-12
Carbine	25-45 m	5-6	3-10
Rifle	30-60 m	5-7	None or 3-10
Battle Rifle	35-70 m	6-7	3-10
Shotgun	10-35 m	5-7	2-5
Heavy Machine Gun	40-100+ m	6-8	10-40

These are construction ranges rather than fixed requirements.

29.2 Accuracy Rating

Accuracy normally ranges from 0 to 3.

It is used only in Burst Effectiveness unless a specific weapon rule says otherwise.

Higher Accuracy represents a weapon that maintains a more useful point of aim during sustained fire.

29.3 Recoil Rating

Recoil is the firearm's resistance-control Difficulty.

Higher Recoil means a greater Strength requirement to control a burst.

Recoil is most important during burst fire and may have little or no effect on single-shot attacks.

29.4 Short Barrel

- Range -5
- Accuracy -1
- Recoil +1
- AP -1
- Weight -1

Useful for compact weapons.

29.5 Standard Barrel

No modifier.

29.6 Long Barrel

- Range +10
- Accuracy +1
- AP +1
- Weight +1

29.7 Heavy Long Barrel

- Range +15
- Damage +1
- Accuracy +1
- Recoil -1
- AP +1
- Weight +2

29.8 Match Barrel

- Range +10
- Accuracy +2
- AP +1
- Weight +1

Best suited to weapons intended for precision or controlled fire.

29.9 Single-Shot Action

Capacity 1 unless another loading system is attached.

No burst capability.

29.10 Bolt Action

- Accuracy +1
- No burst

29.11 Lever Action

- AP 5 baseline
- May cycle one shot between attacks without requiring a separate reload action when the weapon permits it.

29.12 Semi-Automatic Action

Standard action for repeated single shots.

29.13 Automatic Action

Allows burst fire.

Automatic operation may increase Recoil compared with a comparable semi-automatic firearm.

29.14 Closed-Bolt Automatic

- Accuracy +1
- Burst generally 3-8

29.15 Open-Bolt Automatic

- Accuracy -1
- Allows larger bursts
- Recoil generally +1

29.16 No Stock

- Accuracy -1 on burst effectiveness
- Recoil +1
- Weight -1

29.17 Folding Stock

Normal values when extended.

When folded, treat the weapon as having No Stock for purposes of burst effectiveness.

29.18 Fixed Stock

- Recoil -1

29.19 Heavy Stock

- Recoil -1
- Weight +1

29.20 Precision Stock

- Accuracy +1
- Recoil -1
- Weight +1

29.21 Improved Grip

- Recoil -1

29.22 Recoil Grip

- Recoil -1
- Weight +1

29.23 Heavy Fore-End

- Recoil -1
- Weight +1

29.24 Flash Hider

Suppresses visible muzzle flash.

No normal numeric change.

29.25 Compensator

- Recoil -1

29.26 Muzzle Brake

- Recoil -2

The weapon becomes louder and produces a stronger visible signature.

29.27 Suppressor

- Damage -1
- Range -5
- Weight +1

Reduces firing noise but does not make the weapon silent.

29.28 Improved Iron Sights

- Accuracy +1

29.29 Reflex Sight

- Accuracy +1 at short and medium range

29.30 Telescopic Scope

- Accuracy +1
- Range +10
- Targeted attacks cost +1 AP

29.31 Precision Scope

- Accuracy +2
- Range +20
- Targeted attacks cost +1 AP

29.32 Night Sight

Ignores ordinary darkness Difficulty for attacks made through the sight.

29.33 Laser Sight

Targeted attacks ignore the first +2 Difficulty from small body locations.

29.34 Standard Magazine

No modifier.

29.35 Extended Magazine

Capacity becomes:

Base Capacity + floor(Base Capacity / 2)

Reload AP +1.

Weight +1.

29.36 Drum Magazine

Capacity doubles.

Reload AP +2.

Weight +2.

29.37 Reduced Magazine

Capacity is halved, rounded down, with a minimum capacity of 1.

Weight -1.

Useful where concealment and reduced bulk matter.

29.38 Speed Loader

For compatible revolvers and similar systems:

Reload AP becomes 1.

29.39 Bipod

When deployed and fired from a supported position:

- Recoil -2
- Accuracy +1

29.40 Tripod

When deployed:

- Recoil -4
- Accuracy +2
- Range +20

The weapon cannot be moved normally while deployed.

30. Ammunition

Ammunition remains separate from the firearm so that additional ammunition types can be introduced without rewriting weapon entries.

30.1 Full Metal Jacket / Standard

No modifier.

30.2 Hollow Point

Damage +1.

Poor performance against heavy armor.

30.3 Soft Point

Damage +1.

Best against lightly protected targets.

30.4 Match

Accuracy +1 for firearms that accept the ammunition.

30.5 Subsonic

Damage -1.

Reduced firing signature.

30.6 Tracer

No direct damage modifier.

Makes the firing position easier to identify.

30.7 Armor Piercing

PEN 6.

DR is halved, rounded down.

Damage -1.

30.8 Heavy Armor Piercing

PEN 9.

DR is halved, rounded down.

No damage penalty.

30.9 Advanced Armor Piercing

PEN 12.

DR is reduced to one-quarter of normal, rounded down.

Damage -1 unless a specific ammunition profile says otherwise.

30.10 Incendiary

Adds an Environmental fire effect when appropriate.

30.11 Non-Lethal

Damage and effect are defined by the ammunition profile rather than the normal firearm damage.

30.12 Shotgun Ammunition

Shotguns may use multiple ammunition profiles.

Buckshot favors close range and multiple-hit potential.

Slug ammunition concentrates damage into a single projectile.

Specialty shells may produce Environmental or other effects.

31. Armor

Armor uses the same four damage categories used by weapons.

A normal armor profile contains:

Armor	AC	Normal DT / DR	Ballistic DT / DR	Energy DT / DR	Environmental DT / DR
Light Armor	1	2 / 10%	2 / 10%	1 / 5%	1 / 10%
Medium Armor	2	3 / 15%	5 / 25%	2 / 10%	2 / 15%
Heavy Armor	1	6 / 25%	8 / 35%	4 / 20%	4 / 25%
Protective Suit	0	2 / 10%	2 / 10%	2 / 10%	8 / 50%

These are generic examples, not required universal values.

31.1 Environmental Protection

Environmental protection deliberately covers several related hazards at once.

A fully sealed suit may therefore protect against radiation, fire, gas, toxic atmospheres, chemical exposure, and similar threats without requiring a separate armor column for every effect.

A particular hazard may override the normal category when its fictional behavior demands it.

31.2 Armor Penetration and DR

PEN is applied before DR is calculated.

When ammunition halves DR, halve the armor's listed DR after any other applicable DR modifiers and round down.

32. Consumables and Medical Equipment

Consumables should state:

- AP cost to use
- target or area
- immediate effect
- duration
- risks or limitations

Medical items may restore HP, stabilize injuries, remove conditions, or support Doctor and First Aid checks.

PART VI - GAMEMASTER ENGINE

33. NPCs and Adversaries

NPCs do not need full player-style character sheets unless the campaign benefits from them.

A useful NPC profile contains:

- SPECIAL
- important Skills

- HP
- AP
- AC
- relevant resistances
- weapons
- armor
- one or two distinctive abilities

The goal is to have enough information to run the opponent, not to reproduce every player subsystem.

34. Encounter Difficulty

Difficulty should be based on the actual situation rather than character level alone.

Use these rough capability bands when creating opponents:

Skill	Capability
0-4	Untrained
5-9	Basic
10-14	Competent
15-19	Strong
20-24	Exceptional
25-30	Masterful

These are calibration bands, not encounter levels.

A poorly positioned expert can lose to an ordinary opponent with the right weapon, terrain, or surprise.

35. Converting and Customizing the System

The engine is designed to be reskinned without changing the core mathematics.

35.1 Skills

A setting may rename, remove, merge, or add Skills.

A western campaign might replace some technical skills with Riding and Tracking.

A medieval campaign might separate different weapon skills or add Animal Handling.

A science-fiction campaign might split Piloting into Ground, Air, and Space.

A mecha campaign might add Pilot and Gunnery as dedicated skills while leaving the underlying resolution system unchanged.

35.2 SPECIAL

The seven SPECIAL attributes remain useful in almost any setting because they describe broad physical, mental, social, and random capabilities.

35.3 Damage Types

Keep the four broad categories unless the setting truly needs more.

Add a new damage category only when an entire family of weapons, armor, and defenses cannot be expressed cleanly through Normal, Ballistic, Energy, or Environmental.

35.4 Action Points

AP can represent cinematic action economy, tactical positioning, or the number of meaningful things a character can accomplish during a short combat turn.

A high-Agility character should generally receive more AP, while slow characters should have fewer but may still perform expensive actions.

35.5 Firearms

The firearm construction system is intentionally modular.

A GM may build a weapon from components instead of writing unique rules for every model.

The same method can be used for modern firearms, retro-futurist firearms, science-fiction projectile weapons, or other ranged weapons where recoil, accuracy, range, capacity, and firing modes matter.

35.6 Mecha

For a mecha game, treat each machine as equipment or as a character-scale combatant.

Pilot Skill can determine whether the machine does what the pilot intends.

Machine attributes can modify AP, armor, weapons, sensors, and movement.

Damage categories can be retained while adding machine-specific hit locations.

35.7 Medieval Campaigns

For a medieval campaign, replace firearm Skills and equipment with weapon disciplines, riding, mounted combat, shield use, siege operation, or magic as appropriate.

The underlying d20 Skill Test, d10 Attribute Check, AP, armor, and critical systems can remain unchanged.

APPENDIX A - QUICK REFERENCE

Character Creation

1. Roll 1d10 seven times for SPECIAL.
2. Choose a Background and apply exactly +5 SPECIAL points in the Background's chosen distribution.
3. Spend +5 free SPECIAL points.
4. Choose Traits.
5. Choose three Tag Skills.
6. Calculate starting Skills using Appendix B.
7. Apply Background, Trait, and Tag adjustments.
8. Calculate HP, AP, AC, Carry Weight, Melee Damage, Sequence, Poison Resistance, and Radiation Resistance.
9. Choose equipment and finish the character.

Combat Conditions

At 0 HP: Downed.

Bleeding: lose 1 HP at the end of each turn until treated.

Crippled: the affected body part suffers its listed functional penalty until treated.

Dead: only lethal effects, immediately fatal injuries, or the campaign death threshold cause death.

Skill Test

Skill - Difficulty = Success Number

Roll 1d20.

Equal to or below the Success Number succeeds.

Attribute Check

Roll 1d10.

Equal to or below the Attribute succeeds.

Natural 10 always fails.

Advancement

Rule	Value
Skill Points per level	5 + floor(IN / 2)
Skill 1-10	1 point per +1
Skill 11-20	2 points per +1
Skill 21-25	3 points per +1
Skill 26-30	4 points per +1
Tag Skill creation bonus	+4
Tag Skill advancement	Half normal cost, rounded up
Standard Perk interval	Every 3 levels

Secondary Statistics

Statistic	Formula
Level 1 HP	15 + STR + (2 x EN)
Later HP / level	3 + floor(EN / 2)
AP	5 + floor(AG / 2)
Carry Weight	25 + 25 x STR lb
Melee Damage	STR - 5
Poison Resistance	floor(EN / 2)
Radiation Resistance	floor(EN / 5)
Sequence	2 x PE
Attribute Modifier	Attribute - 5
AC	Armor AC + floor(AG / 2) + other modifiers

Burst Fire

1. Declare burst size.
2. Declare target zone.
3. Declare Brace.
4. Make the normal attack roll.
5. Calculate Attack Margin.
6. Reduce Recoil from firearm skill: 2 for every full 5 Skill above 5.
7. Calculate Recoil Target: Effective STR - Effective Recoil.
8. Roll 1d10 against Recoil Target; natural 10 fails.
9. Calculate Control Margin on success.
10. Calculate Burst Score.
11. Apply target-zone requirement.

12. Use Burst Hit Table.
13. Roll damage once and multiply by the number of hits (Fast Burst). Use the Per-Bullet Burst module only when individual hits must be resolved.

Combat Procedures

Default hit location: Center Mass / Torso.

First Aid: 6 AP in combat; successful First Aid restores 1d10 HP and ends Bleeding.

Surprise: unsurprised characters act first in Sequence order; surprised characters act afterward in Sequence order for the opening round only.

Defend: unused AP adds to AC until the next turn, maximum +4.

PEN / DT / DR

Adjusted DT = max(0, DT - PEN)

Then apply DR to the damage remaining after DT.

Criticals

Luck	Base Critical Range
1-5	1
6-10	1-2
11-15	1-3
16-20	1-4
21-25	1-5
26-30	1-6

Finesse expands the Critical Range by 1.

Jinxed changes Critical Failure to 18-20 for the character and every combatant in the fight.

APPENDIX B - STARTING SKILL FORMULA SHEET

These formulas are direct d20-scale starting relationships. A campaign may rename or replace Skills while keeping the same formula structure.

Round the final value to the nearest whole number. Apply Background, Trait, and Tag bonuses after calculating base Skills.

Skill	Formula
Small Guns	$\text{round}(1 + 0.8 \times \text{AG})$
Big Guns	$\text{round}(0.4 \times \text{AG})$
Energy Weapons	$\text{round}(0.4 \times \text{AG})$
Unarmed	$\text{round}(6 + 0.4 \times (\text{AG} + \text{STR}))$
Melee Weapons	$\text{round}(4 + 0.4 \times (\text{AG} + \text{STR}))$
Throwing	$\text{round}(0.8 \times \text{AG})$
First Aid	$\text{round}(0.4 \times (\text{PE} + \text{EN}))$
Doctor	$\text{round}(1 + 0.2 \times (\text{PE} + \text{IN}))$
Sneak	$\text{round}(1 + 0.6 \times \text{AG})$
Lockpick	$\text{round}(2 + 0.2 \times (\text{PE} + \text{AG}))$
Steal	$\text{round}(0.6 \times \text{AG})$

Skill	Formula
Traps	$\text{round}(2 + 0.2 \times (\text{PE} + \text{AG}))$
Science	$\text{round}(0.8 \times \text{IN})$
Repair	$\text{round}(0.6 \times \text{IN})$
Pilot	$\text{round}(0.4 \times (\text{AG} + \text{PE}))$
Speech	CH
Barter	$\text{round}(0.8 \times \text{CH})$
Gambling	LK
Outdoorsman	$\text{round}(0.4 \times (\text{EN} + \text{IN}))$

APPENDIX C - WORKED CHARACTERS AND COMBAT

C.1 Balanced Generalist

Initial rolls:

STR 6, PE 7, EN 5, CH 6, IN 8, AG 7, LK 6.

Background: Laborer (+3 STR, +1 EN, +1 PE; Melee Weapons +2, Repair +2, Outdoorsman +1).

Free +5: +2 AG, +1 IN, +1 LK, +1 CH.

Final:

STR 9, PE 8, EN 6, CH 7, IN 9, AG 9, LK 7.

$\text{AP} = 5 + \text{floor}(9 / 2) = 9.$

AC with Light Armor (AC 1) = $1 + \text{floor}(9 / 2) = 5.$

Level 1 HP = $15 + 9 + (2 \times 6) = 36.$

C.2 High-Intelligence Specialist

Initial rolls:

STR 4, PE 8, EN 5, CH 5, IN 10, AG 6, LK 8.

Background: Academic (+4 IN, +1 PE; Science +2, Doctor +2, First Aid +1).

Free +5: +1 IN, +2 PE, +1 CH, +1 LK.

Final:

STR 4, PE 11, EN 5, CH 6, IN 15, AG 6, LK 9.

Skill Points per level = $5 + \text{floor}(15 / 2) = 12.$

C.3 Worked Combat

Mira has Small Guns 16, AG 8, STR 7, LK 8, and 8 AP. She carries a pistol dealing $1d10 + 2$ damage. The pistol's single-shot attack costs 4 AP.

Dane has Small Guns 18, AG 8, STR 5, LK 10, and 10 AP. He carries a burst-capable carbine with Burst 5, Accuracy +2, Recoil 8, and damage $1d6 + 2$. He is Braced for the burst.

Dane is behind **Solid Cover**, which imposes +4 Difficulty on attacks against him. Both characters are at normal range and the area is well lit.

Mira's single shot

Dane has Armor AC 2 and AG 8, so $AC = 2 + \text{floor}(8 / 2) = 6$. Mira's Success Number is $16 - 4 \text{ cover} - 6 AC = 6$. Mira rolls **5**, so the attack succeeds. Luck 8 gives a Critical Range of 1-2, so this is not a Critical Hit.

Mira rolls **1d10 + 2 = 8 damage**. Dane's armor is **Ballistic DT 5 / DR 25%**. After DT, $8 - 5 = 3$. DR removes 0, so Dane takes **3 damage**.

Dane's braced burst

Dane fires a 5-round burst. Mira has AC 6, so Dane's Success Number is $18 - 6 = 12$. He rolls a **2**. The attack succeeds and is a Critical Hit because Luck 10 gives a Critical Range of 1-2.

Attack Margin = $12 - 2 = 10$. Small Guns 18 gives Recoil Training -4, so Effective Recoil = $8 - 4 = 4$. Braced STR = $5 \times 2 = 10$, giving a Recoil Target of $10 - 4 = 6$.

Dane rolls **3** on the Recoil Check, giving Control Margin $6 - 3 = 3$.

Burst Score = $(10 \times 2) + 2 + (3 \times 2) = 28$. The Burst Hit Table gives 9 hits, capped by the 5 rounds fired, so **5 hits** land.

Using default Fast Burst, Dane rolls damage once: **1d6 + 2 = 5**. Five hits produce **25 raw damage**. Roll the Torso Critical table; suppose the result is Double Damage. The burst becomes **50 raw damage** before armor.

Mira's armor is **Ballistic DT 8 / DR 35%**. After DT: $50 - 8 = 42$. DR removes $\text{floor}(42 \times 0.35) = 14$. Final damage is **28 HP**.

This example resolves cover, AC, one single shot, Brace, recoil, Burst Score, Fast Burst, DT, DR, and one Critical Hit using the standard procedures.

APPENDIX D - FIREARM CONSTRUCTION REFERENCE

Basic Construction Sequence

1. Select firearm family.
2. Select cartridge from the Cartridge Table.
3. Select action.
4. Select barrel.
5. Select stock.
6. Select sights.
7. Select magazine.
8. Select recoil-control parts.
9. Add suppressor or muzzle device if desired.
10. Determine final Damage, Range, AP, Accuracy, Recoil, Capacity, Weight, and Min STR.
11. Compare the result to the campaign's existing weapons.

Recoil Formula

Recoil Training = $2 \times \text{floor}((\text{Firearm Skill} - 5) / 5)$, minimum 0

Effective Recoil = **Weapon Recoil** - **Recoil Training**, minimum 0

Effective STR = **STR** $\times 2$ while Braced; otherwise **STR**

Recoil Target = **Effective STR** - **Effective Recoil**, minimum 1, maximum 9

Natural 10 always fails the Recoil Check.

Burst Formula

Attack Margin = Success Number - attack die

Control Margin = Recoil Target - recoil die on a successful Recoil Check; otherwise 0

Burst Score = (Attack Margin x 2) + Accuracy + (Control Margin x 2)

Then apply the Target Zone requirement.

Target Zones

Zone	Requirement
Center Mass	+0
Arms / Legs / Groin	+4
Head	+8
Eyes	+12

Burst Hit Table

Effective Burst Score	Hits
0-6	1
7-9	2
10-12	3
13-15	4
16-18	5
19-21	6
22-24	7
25-27	8
28-30	9
31+ Full burst, up to the declared burst size	

Cartridge Table

The values below are engine construction values, not ballistic measurements. Use them as game-balance packages.

Cartridge	Damage	PEN	Recoil Contribution	Range Contribution
.22 LR	1d6	1	-2	-10
.380 ACP	1d8	2	-1	-5
9x19mm	1d8 + 1	3	0	0
.45 ACP	1d10	3	+1	-5
.357 Magnum	1d10 + 1	4	+2	+5
5.56x45mm	1d10 + 2	5	+1	+15
5.45x39mm	1d10 + 1	5	+1	+15
7.62x39mm	2d6	6	+2	+10
7.62x51mm	2d6 + 1	7	+3	+20
.30-06 Springfield	2d6 + 2	8	+4	+25
12 Gauge Buckshot	2d6	3	+4	-10
12 Gauge Slug	2d10	6	+5	+5

Cartridge modifiers are added to the firearm's component profile. Recoil and Range are numeric construction values; Accuracy and AP remain properties of the assembled firearm unless a component changes them.

Common Component Effects

Component	Damage	Accuracy	Recoil	Range	AP	Weight
Short Barrel	-1	-1	+1	-5	-1	-1
Long Barrel	0	+1	0	+10	+1	+1
Heavy Long Barrel	+1	+1	-1	+15	+1	+2
Match Barrel	0	+2	0	+10	+1	+1
No Stock	0	-1	+1	0	0	-1
Fixed Stock	0	0	-1	0	0	0
Heavy Stock	0	0	-1	0	0	+1
Precision Stock	0	+1	-1	0	0	+1
Compensator	0	0	-1	0	0	0
Muzzle Brake	0	0	-2	0	0	0
Suppressor	-1	0	0	-5	0	+1
Improved Sights	0	+1	0	0	0	0
Telescopic Scope	0	+1	0	+10	0	+1
Precision Scope	0	+2	0	+20	0	+1
Bipod	0	+1	-2	0	0	+1
Tripod	0	+2	-4	+20	0	+7